ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 2 · PRACTICE C · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals paper 2 · practice c.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Practical work is not marked strictly on syntax. Give marks for correct logic, correct pins and connections, sensible comments and neat work. Accept other pin choices and layouts that work.

QUESTION 1

1.1 Six LEDs placed in a row ✓; each anode connected to its own GPIO pin ✓✓ through its own 220 Ω resistor ✓✓; all cathodes to the GND rail ✓; GND rail connected to a Raspberry Pi GND pin ✓; LED polarity correct ✓; neat, no short circuits ✓; pins labelled / correct GPIO numbers ✓ (10)

1.2 Start ✓; set up the six GPIO pins as outputs (list of pins) ✓✓; loop forever ✓; for each LED in order (left to right) ✓: switch it on ✓, wait 0,3 s ✓, switch it off ✓; after the sixth LED go back to the first ✓; on Ctrl + C clean up the GPIO pins ✓ and stop ✓ (11)

1.3 Circuit saved and printed / presented ✓; flowchart saved and printed / presented ✓ (2)

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QUESTION 2

2.1 File created and named correctly ✓✓ (2)

2.2 (1) BCM ✓ (2) IN ✓ (3) OUT ✓ (4) True ✓ (5) HIGH ✓ (6) HIGH ✓ (7) LOW ✓ (8) LOW ✓ (9) sleep ✓ (10) KeyboardInterrupt ✓ (11) cleanup ✓ (11)

2.3 Useful comments, about 1 mark each ✓✓✓✓✓✓✓✓✓✓, e.g. importing the libraries; pin numbers of the sensor, LED and buzzer; BCM numbering; sensor as input; LED and buzzer as outputs; loop that keeps checking; dry-soil decision; beeping the buzzer; moist soil switches the LED off; waiting 5 s between checks; Ctrl + C handling; cleaning up the pins. (10)

2.4 Saved ✓ and printed / presented ✓ (2)

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QUESTION 3

3.1 Flame sensor VCC to 3,3 V ✓, GND to GND ✓, DO to a GPIO pin ✓; red LED with 220 Ω to a GPIO pin ✓; green LED with 220 Ω to a GPIO pin ✓; buzzer to a GPIO pin and GND ✓; push button between a GPIO pin and GND (internal pull-up) ✓; LED polarities correct ✓; common GND ✓; neat, labelled ✓ (10)

3.2 File created and named ✓✓ (2)

3.3 Libraries imported and pins set up ✓✓; green LED on while there is no flame ✓; flame detected when DO is LOW ✓; alarm latched (variable) so it continues after the flame disappears ✓; red LED flashes ✓ and buzzer sounds ✓; green LED off during the alarm ✓; RESET button stops the alarm ✓; try / except / finally with cleanup ✓, e.g. (10)

```
import RPi.GPIO as GPIO
import time
```

```
FLAME = 5      # flame sensor DO, LOW = flame
RED = 6
GREEN = 13
BUZZER = 19
RESET = 26     # push button to GND

GPIO.setmode(GPIO.BCM)
GPIO.setup(FLAME, GPIO.IN)
GPIO.setup(RESET, GPIO.IN, pull_up_down=GPIO.PUD_UP)
GPIO.setup([RED, GREEN, BUZZER], GPIO.OUT)

alarm = False
try:
    while True:
        if GPIO.input(FLAME) == GPIO.LOW:      # flame detected
            alarm = True
        if GPIO.input(RESET) == GPIO.LOW:      # teacher pressed RESET
            alarm = False
        if alarm:
            GPIO.output(GREEN, GPIO.LOW)
            GPIO.output(BUZZER, GPIO.HIGH)
            GPIO.output(RED, GPIO.HIGH)        # flash the red LED
            time.sleep(0.3)
            GPIO.output(RED, GPIO.LOW)
            time.sleep(0.3)
        else:
            GPIO.output(BUZZER, GPIO.LOW)
            GPIO.output(RED, GPIO.LOW)
            GPIO.output(GREEN, GPIO.HIGH)      # system OK
            time.sleep(0.1)
except KeyboardInterrupt:
    print("Fire alarm stopped")
finally:
    GPIO.cleanup()
```

3.4 Name and description of the system ✓✓; basic operation – how to switch on, what the green / red LEDs and buzzer mean, how to reset ✓✓✓; troubleshooting – at least two problems with solutions (e.g. green LED off: check power and wiring; false alarms: adjust the sensor's sensitivity screw or move it away from sunlight) ✓✓✓; at least two sensible FAQs with answers ✓✓ (10)

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TOTAL: 80

END OF MARKING GUIDELINE